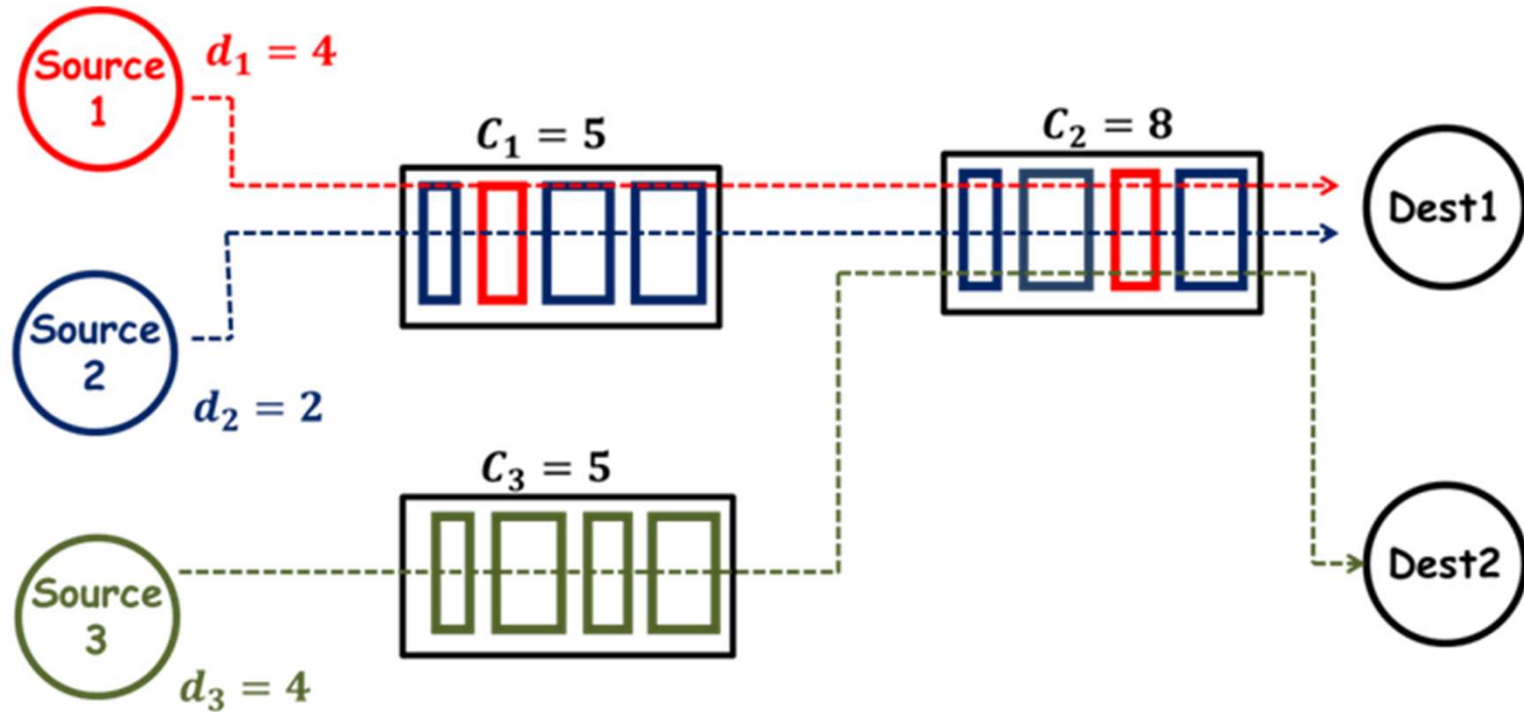


Tutorial 4

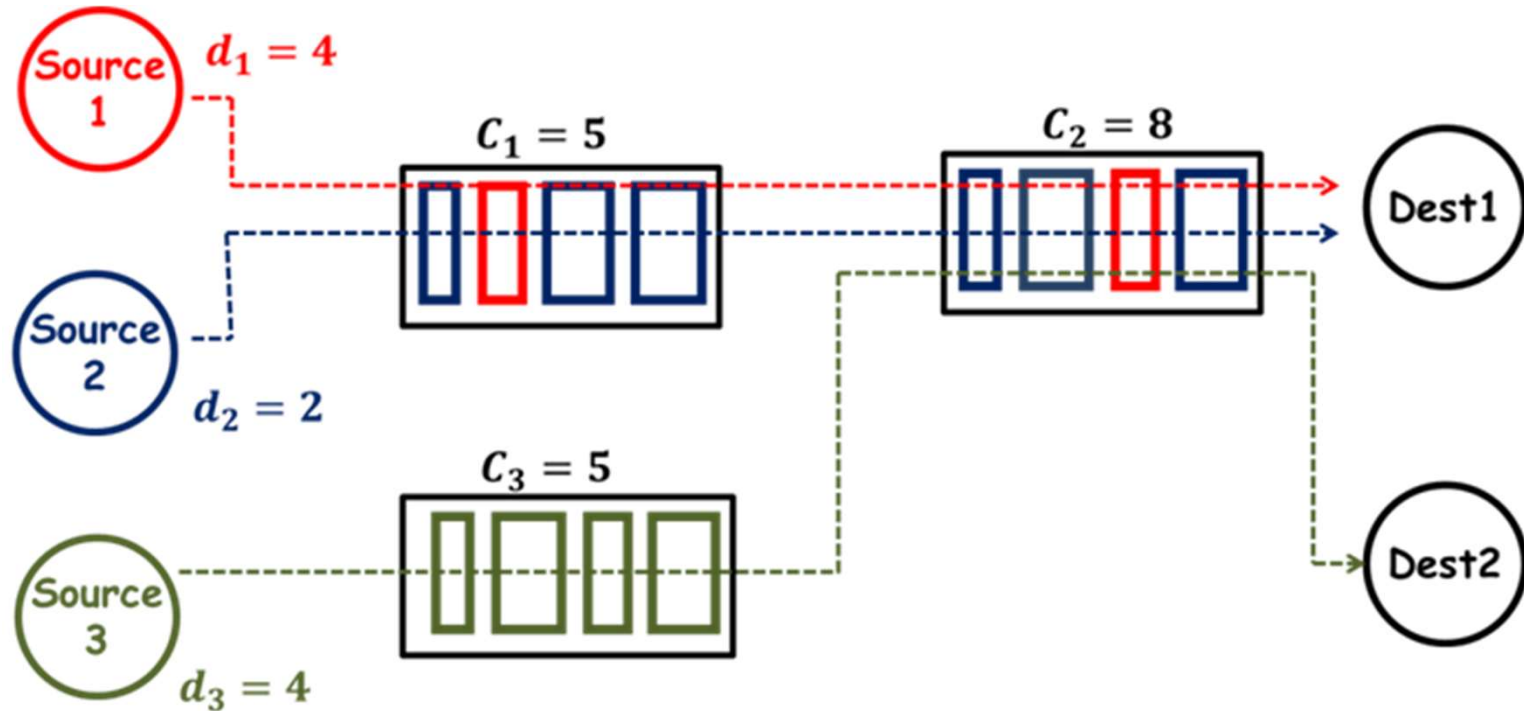
CS4226: Internet Architecture

Q1



- Please identify the maxmin fair allocation for the three flows and the bottleneck link(s).

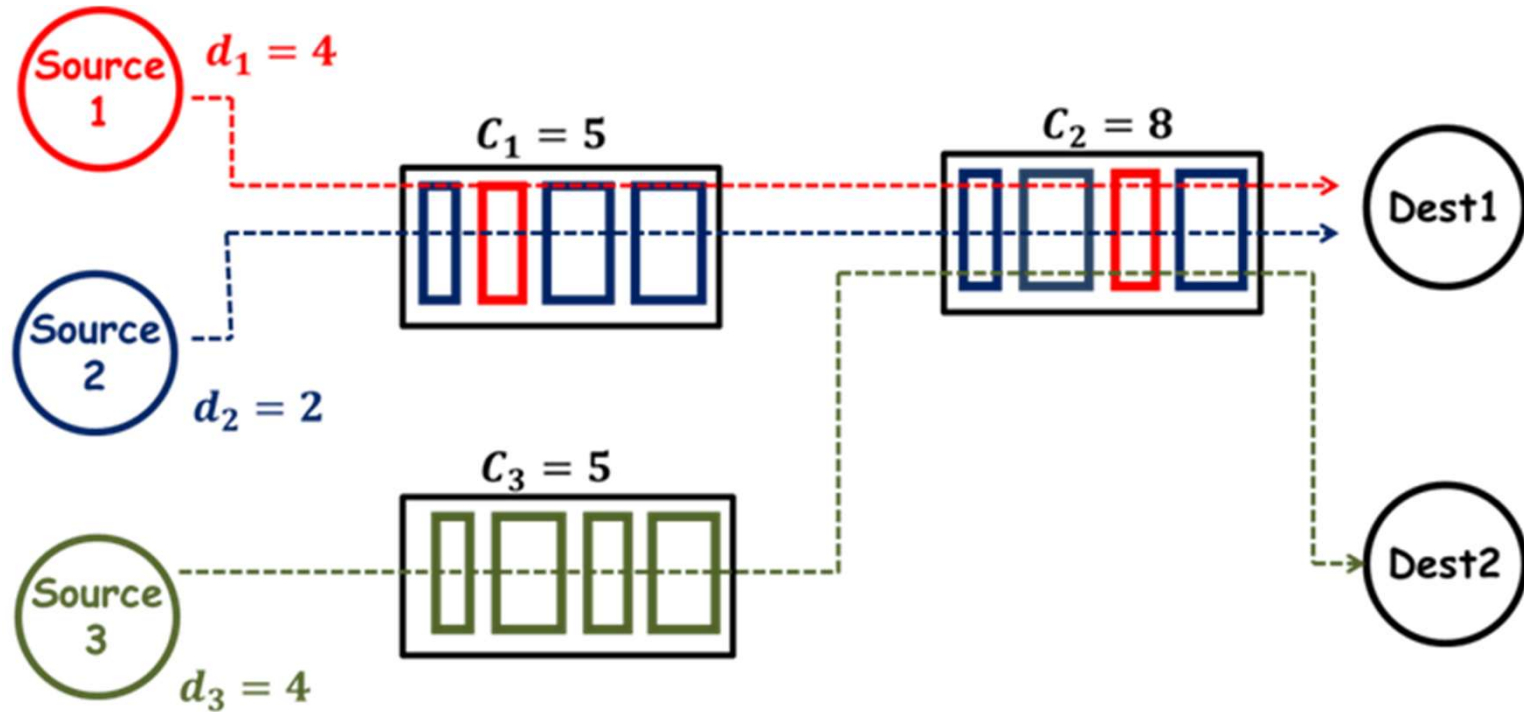
Q1



□ Some observations:

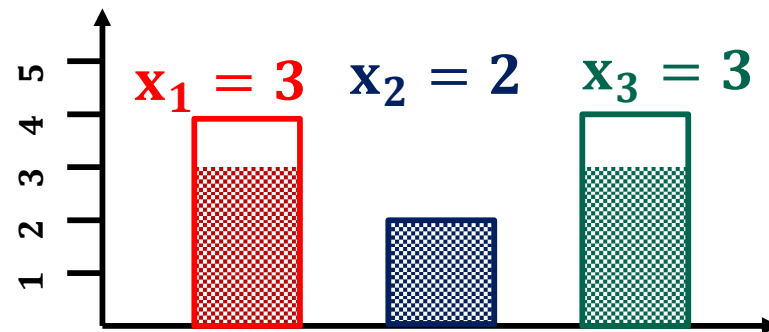
- ❖ C_3 is not a bottleneck of flow 3
- ❖ flow 2 needs less than its fair share at both C_1 and C_2 .
- $x_2 = d_2 = 2$
- $x_1 = x_3 = 3$ (at C_2)

Q1

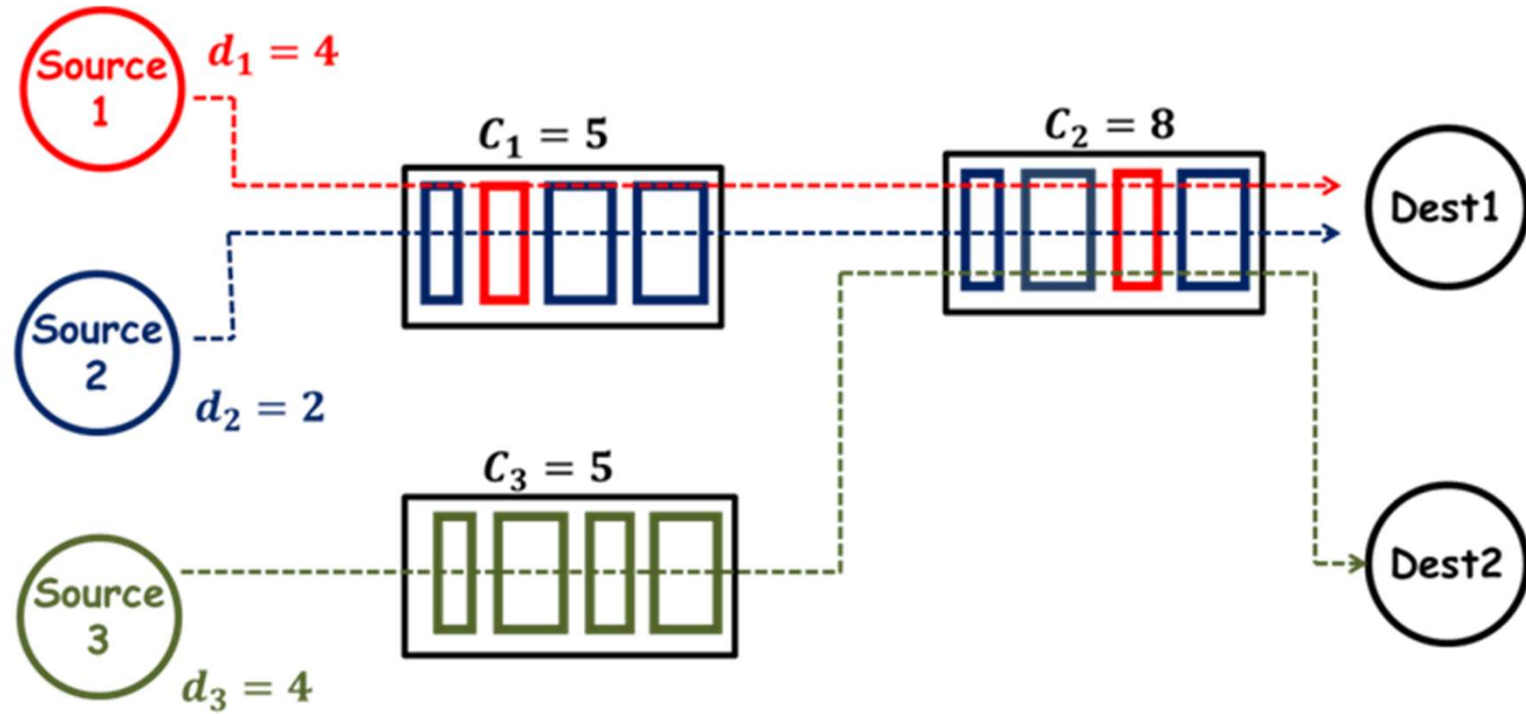


□ How about water-filling?

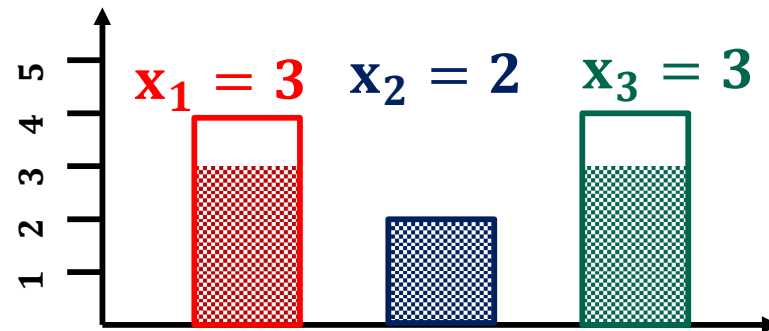
- ❖ $x_1 + x_2 \leq 5$
- ❖ $x_1 + x_2 + x_3 \leq 8$
- ❖ $x_3 \leq 5$



Q1



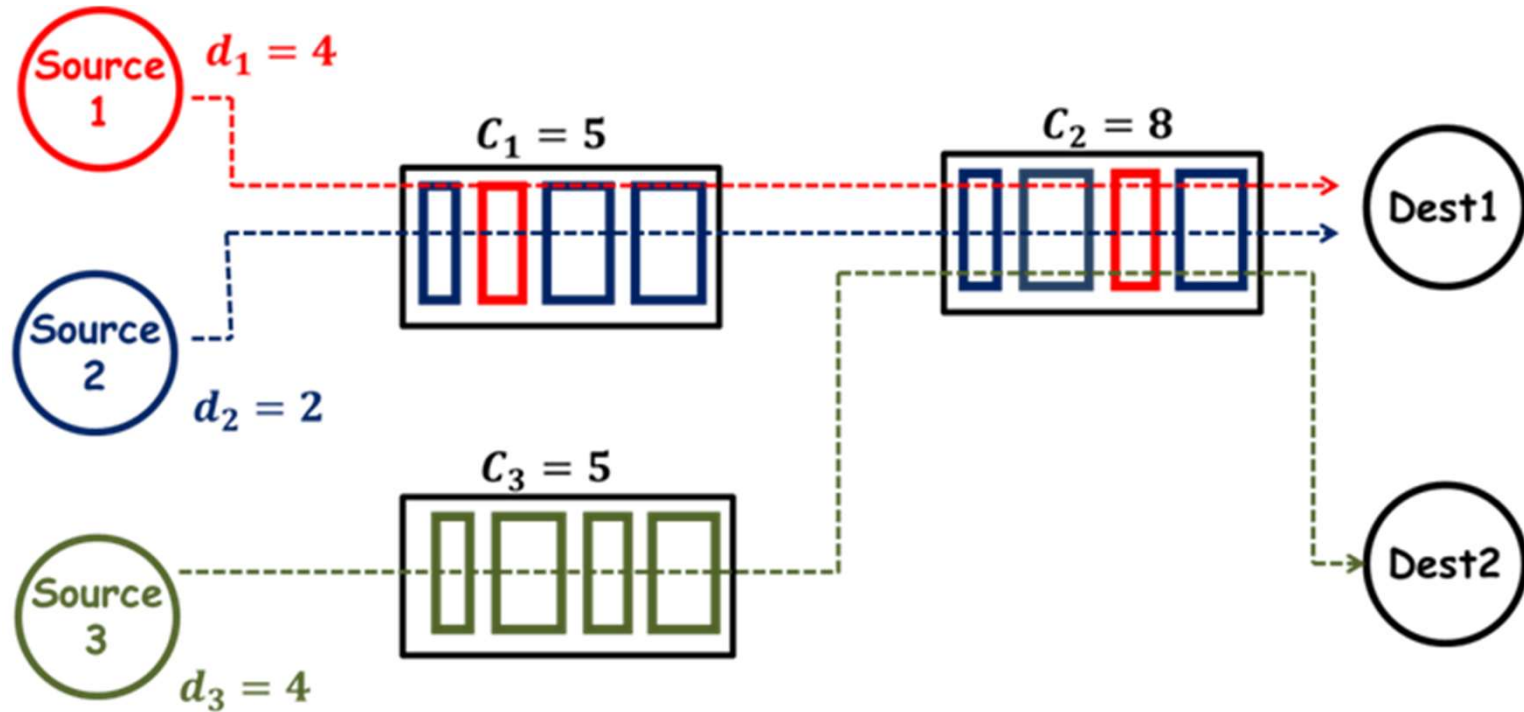
□ How about bottleneck?



Bottleneck definition

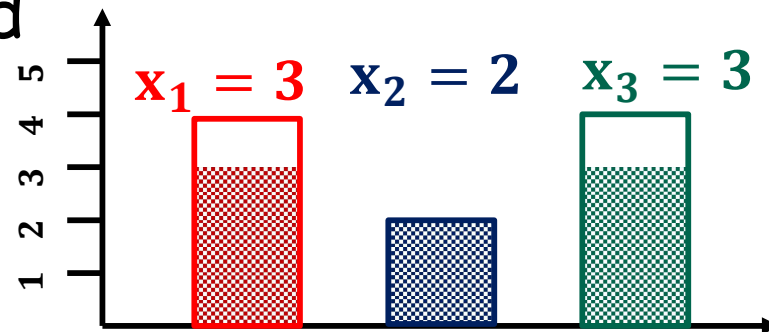
- In the network case, we say that resource r is a bottleneck for flow i if and only if
 - resource r is saturated and
 - flow i has the maximum rate among all flows using resource r .

Q1

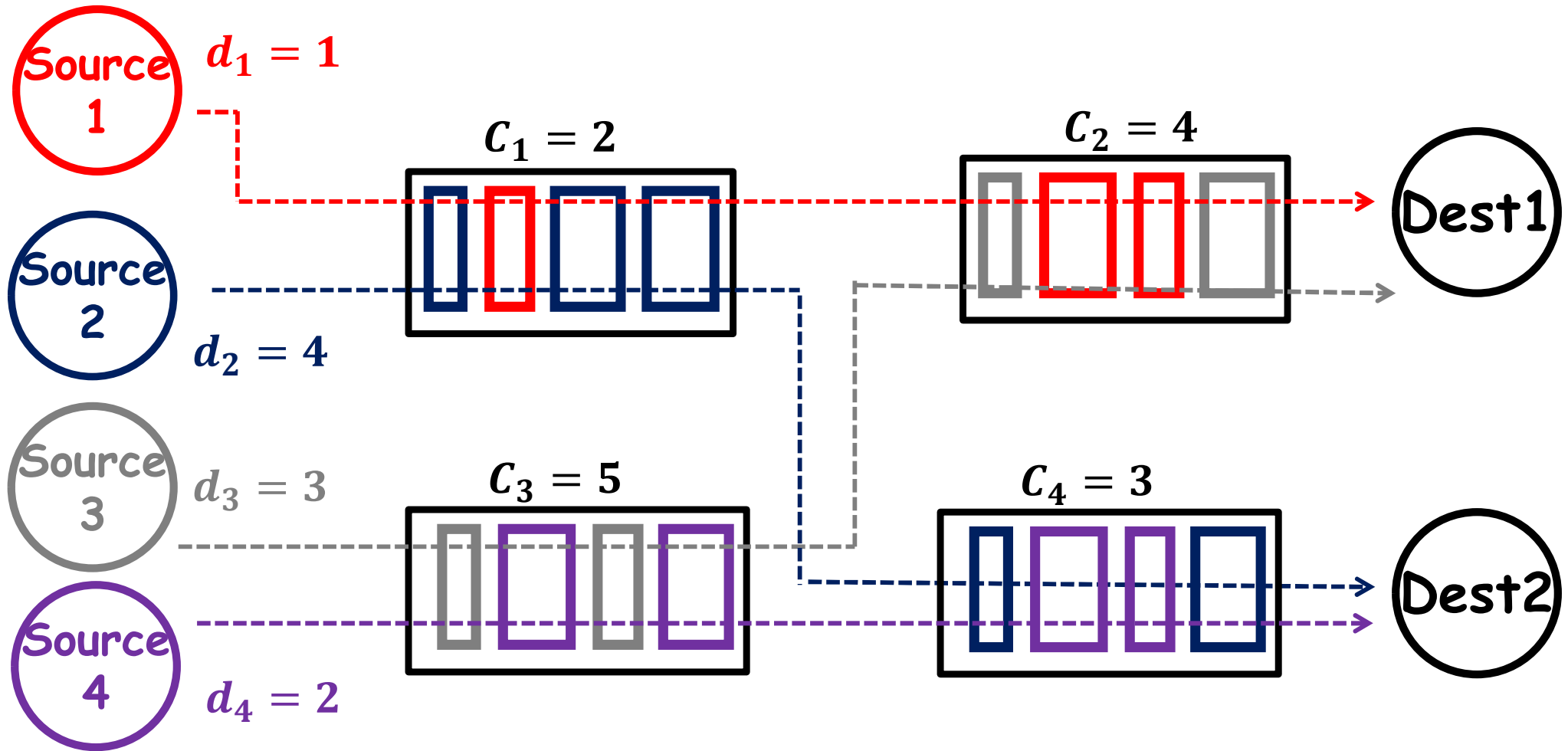


□ How about bottleneck?

- ❖ C_1 and C_2 are saturated
- ❖ flow 3 gets the max throughput at C_2
- ❖ flow 1 gets the max throughput at C_1 and C_2

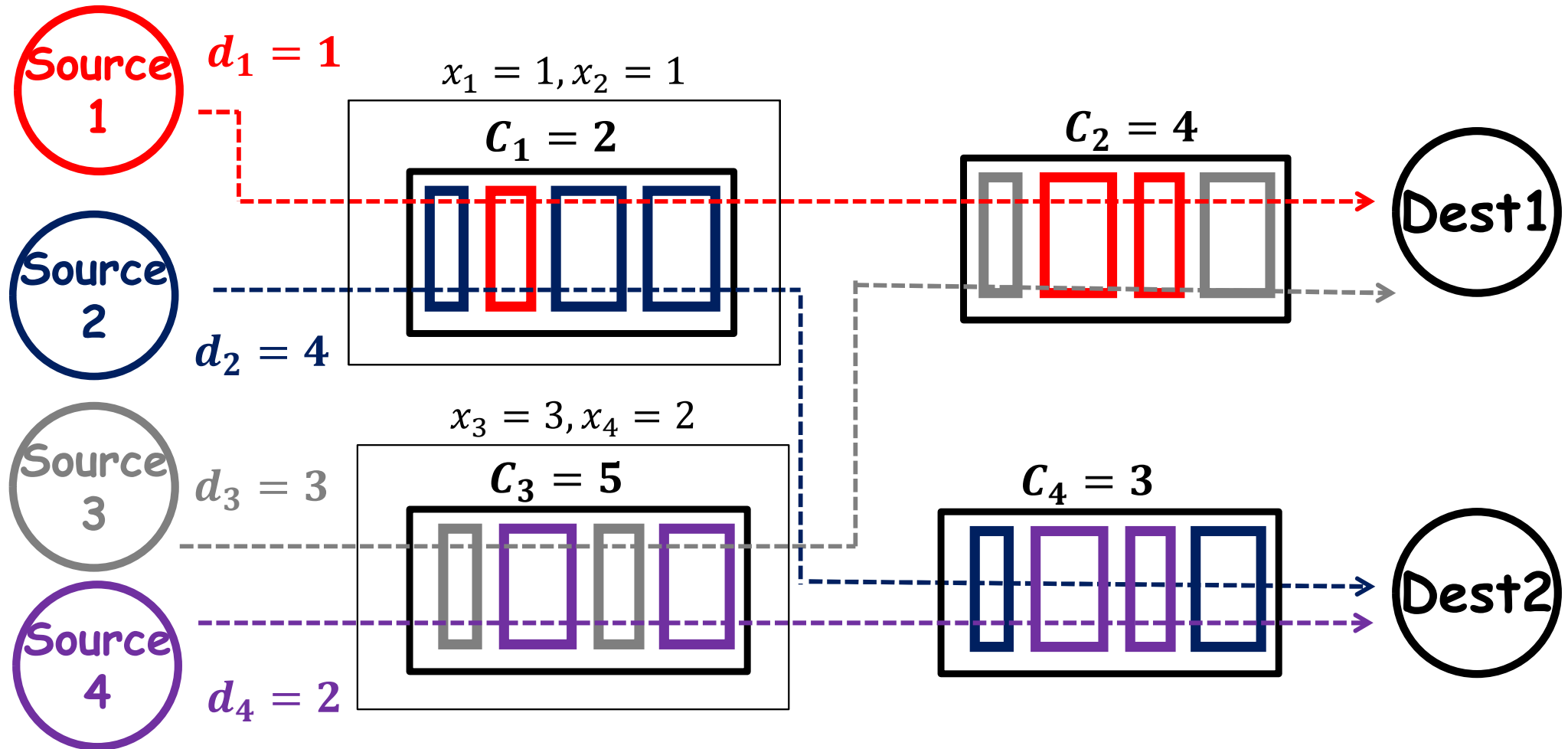


Q2.A



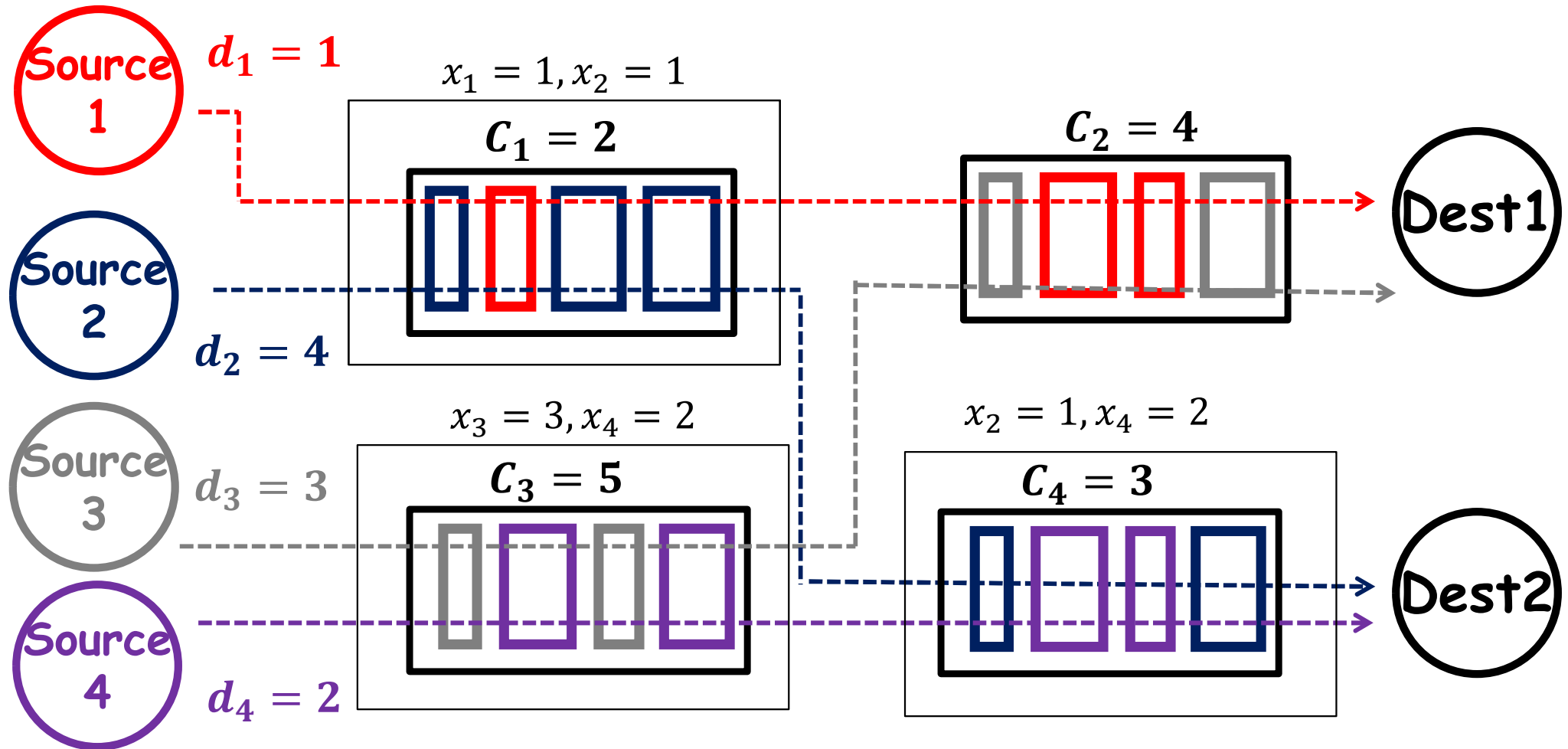
□ Calculate the max-min fair solution:

Q2.A



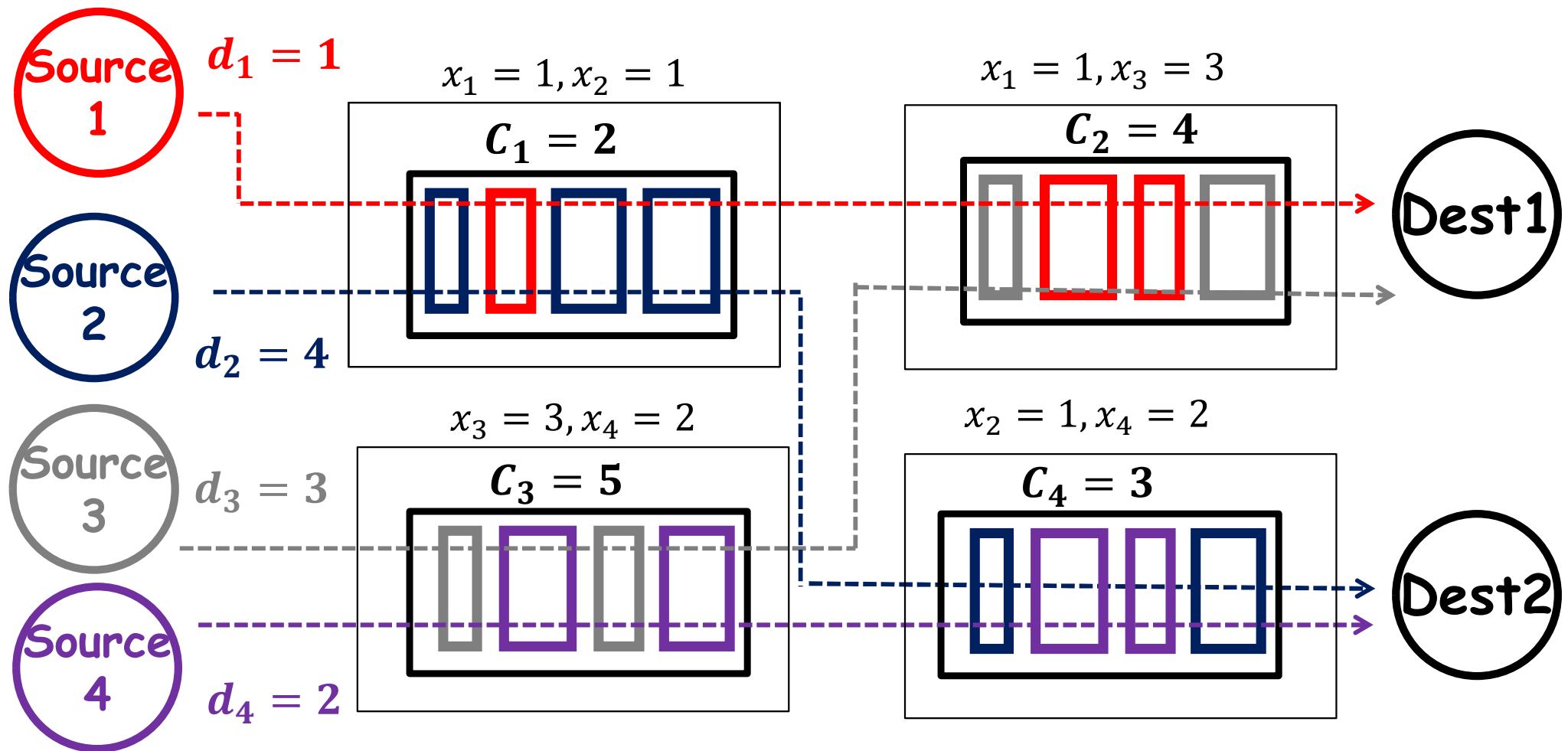
□ Calculate the max-min fair solution:

Q2.A



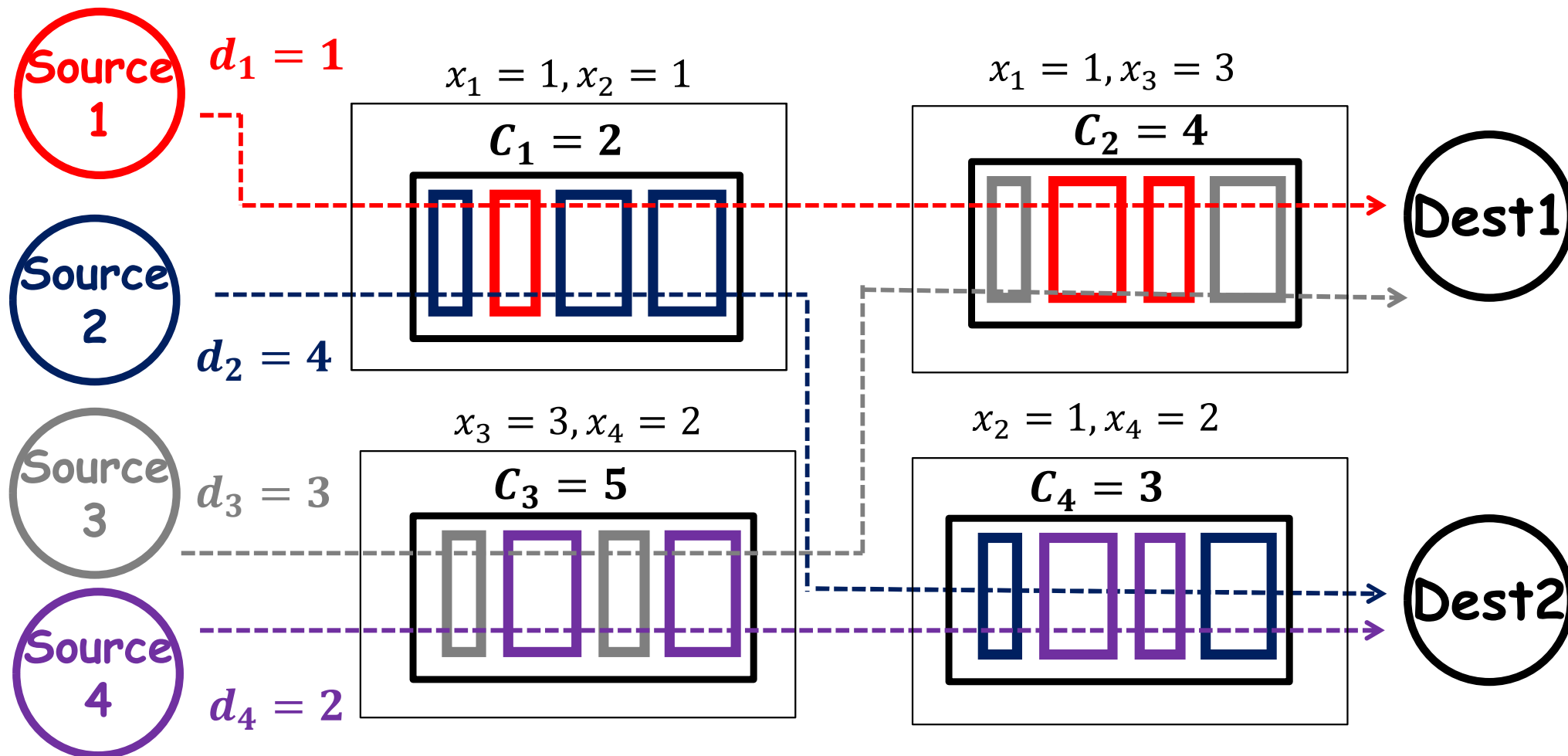
□ Calculate the max-min fair solution:

Q2.A



□ Calculate the max-min fair solution:

Q2.A

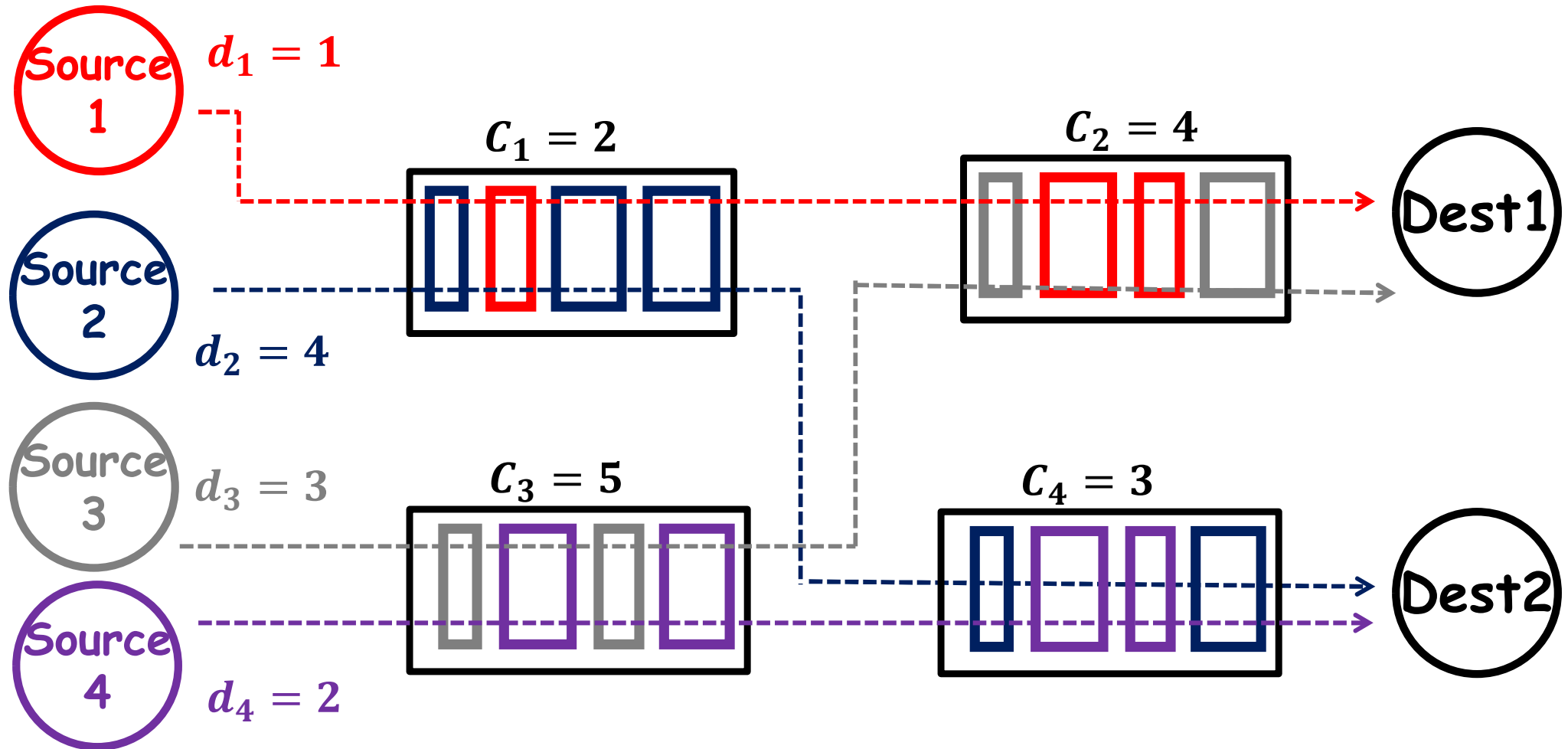


□ Calculate the max-min fair solution:

$$x_1 = x_2 = 1$$

$$x_3 = 3; x_4 = 2$$

Q2.A

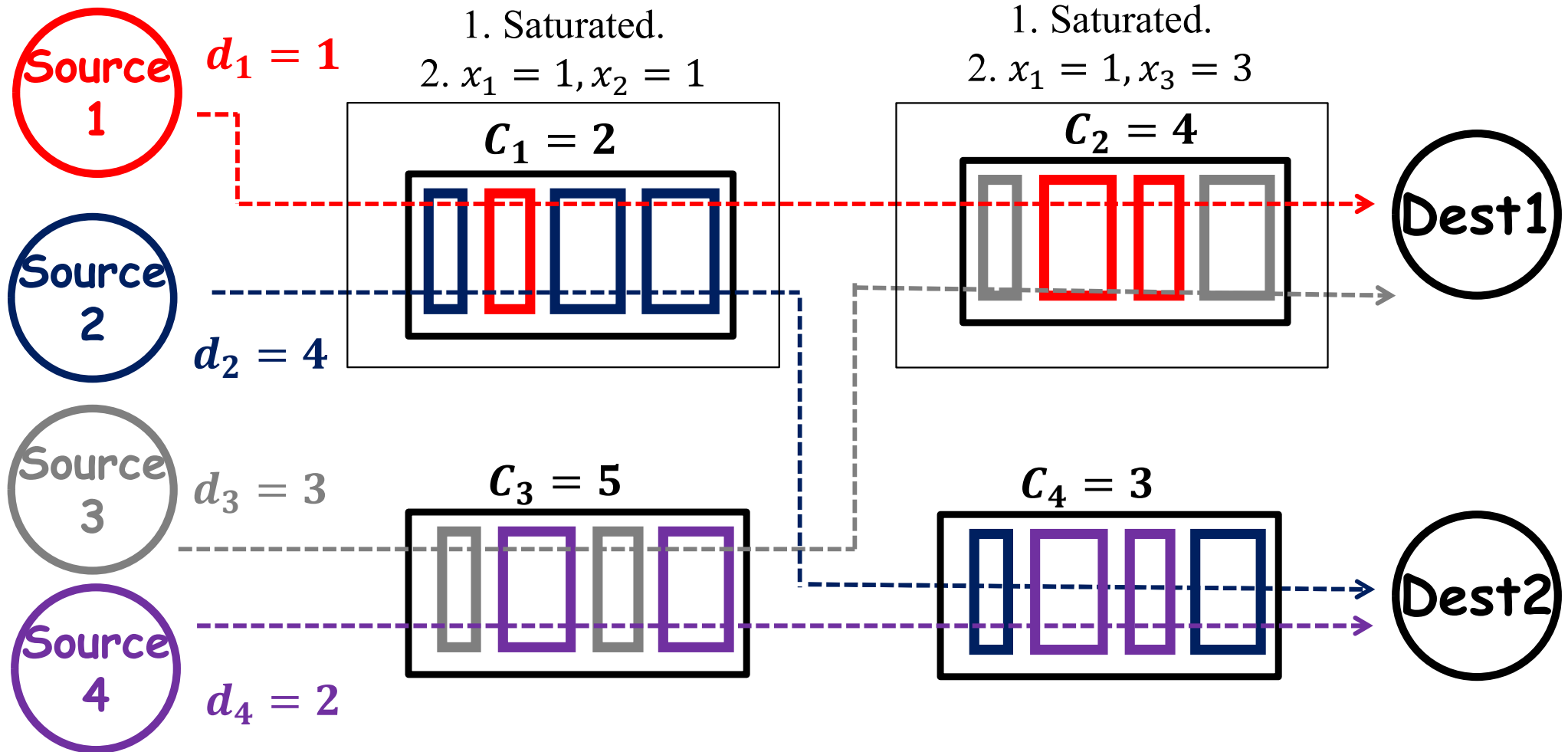


□ Bottleneck links for each flow:

Bottleneck definition

- In the network case, we say that resource r is a bottleneck for flow i if and only if
 - resource r is saturated and
 - flow i has the maximum rate among all flows using resource r .

Q2.A



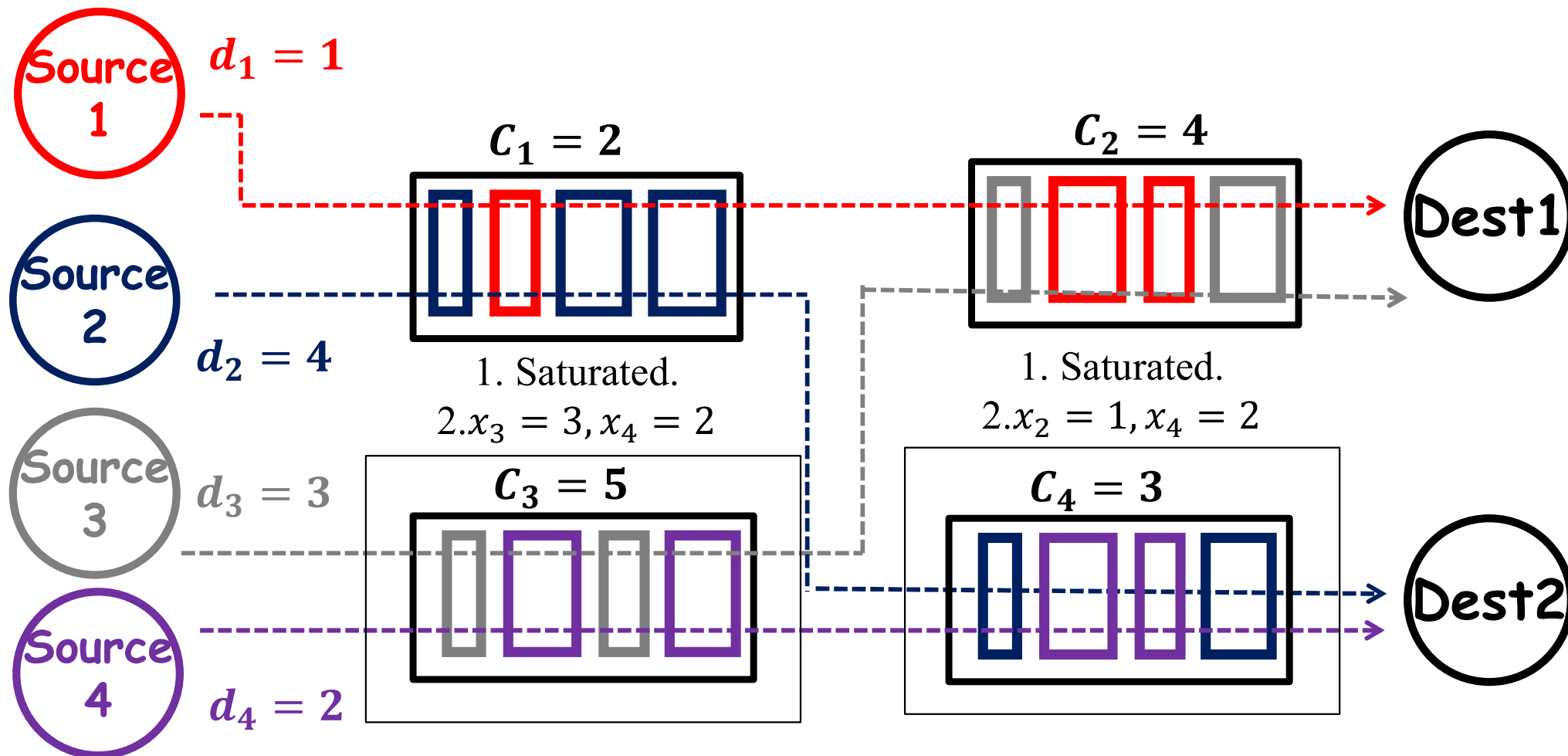
□ Bottleneck links for each flow:

d_1 d_2 is bottlenecked by C_1

d_3 is bottlenecked by C_2

Even the flow has been fulfilled!

Q2.A

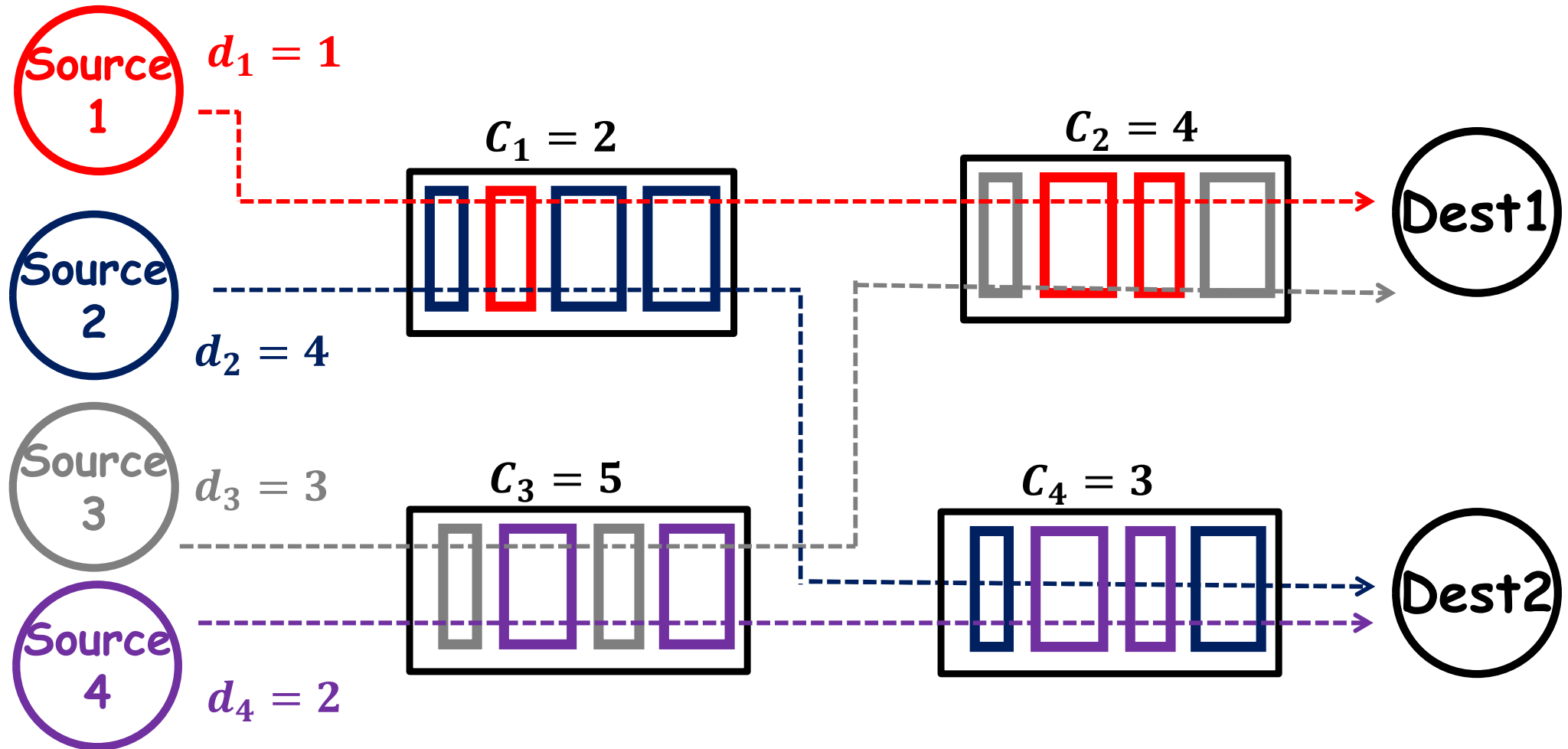


□ Bottleneck links for each flow:

- d_1 d_2 is bottlenecked by C_1
- d_3 is bottlenecked by C_2, C_3
- d_4 is bottlenecked by C_4

$$\Phi = (\phi_1, \phi_2, \phi_3, \phi_4) = (2, 3, 4, 5)$$

Q2.B



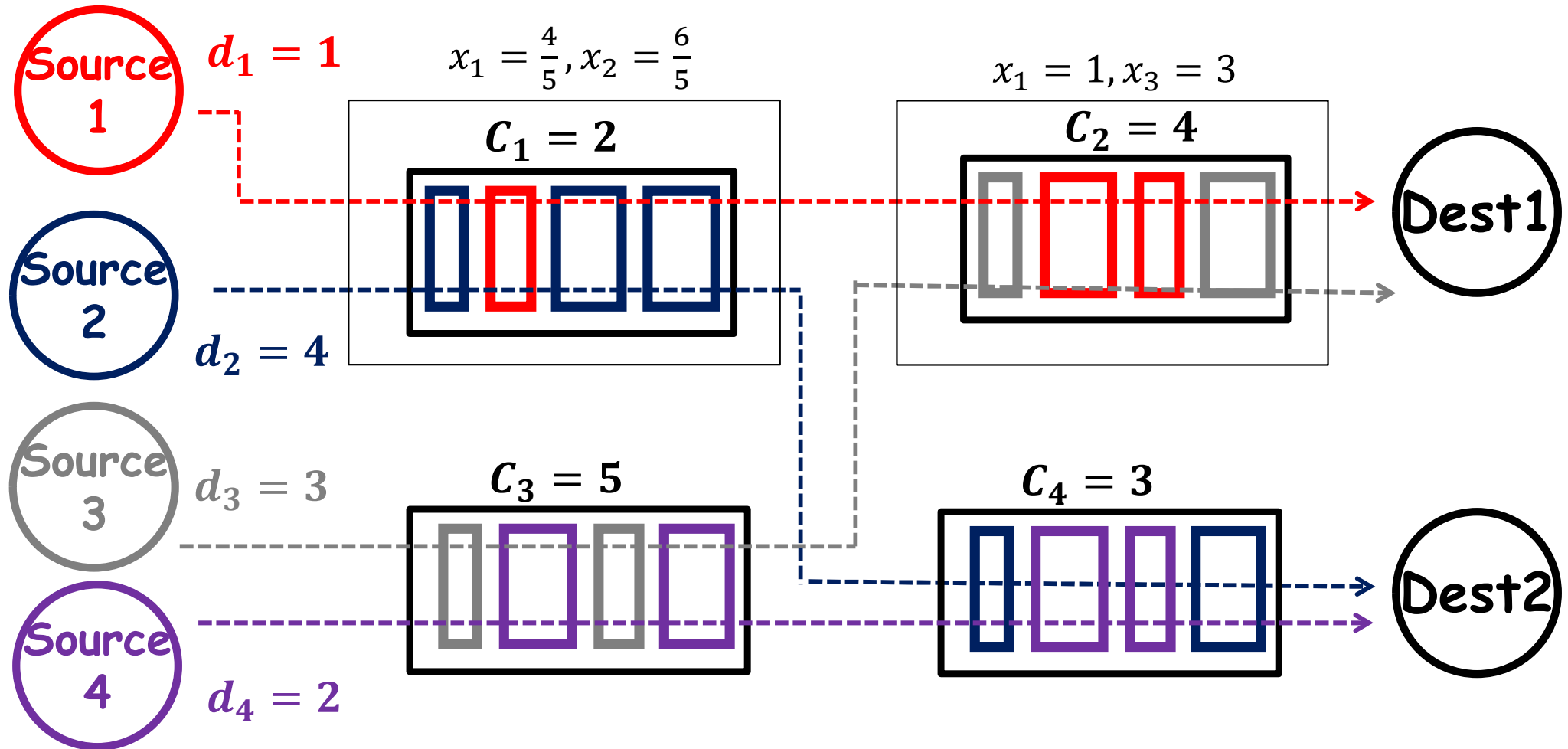
□ Calculate the max-min fair solution:

$$x_i = C_n \phi_i / (\sum \phi_{i \in C_n})$$

$$\Phi = (\phi_1, \phi_2, \phi_3, \phi_4) = (2, 3, 4, 5)$$

Q2.B

d_1 might be fulfilled by C_1 ($x_1 = 4/5$) and C_2 ($x_1 = 1$)



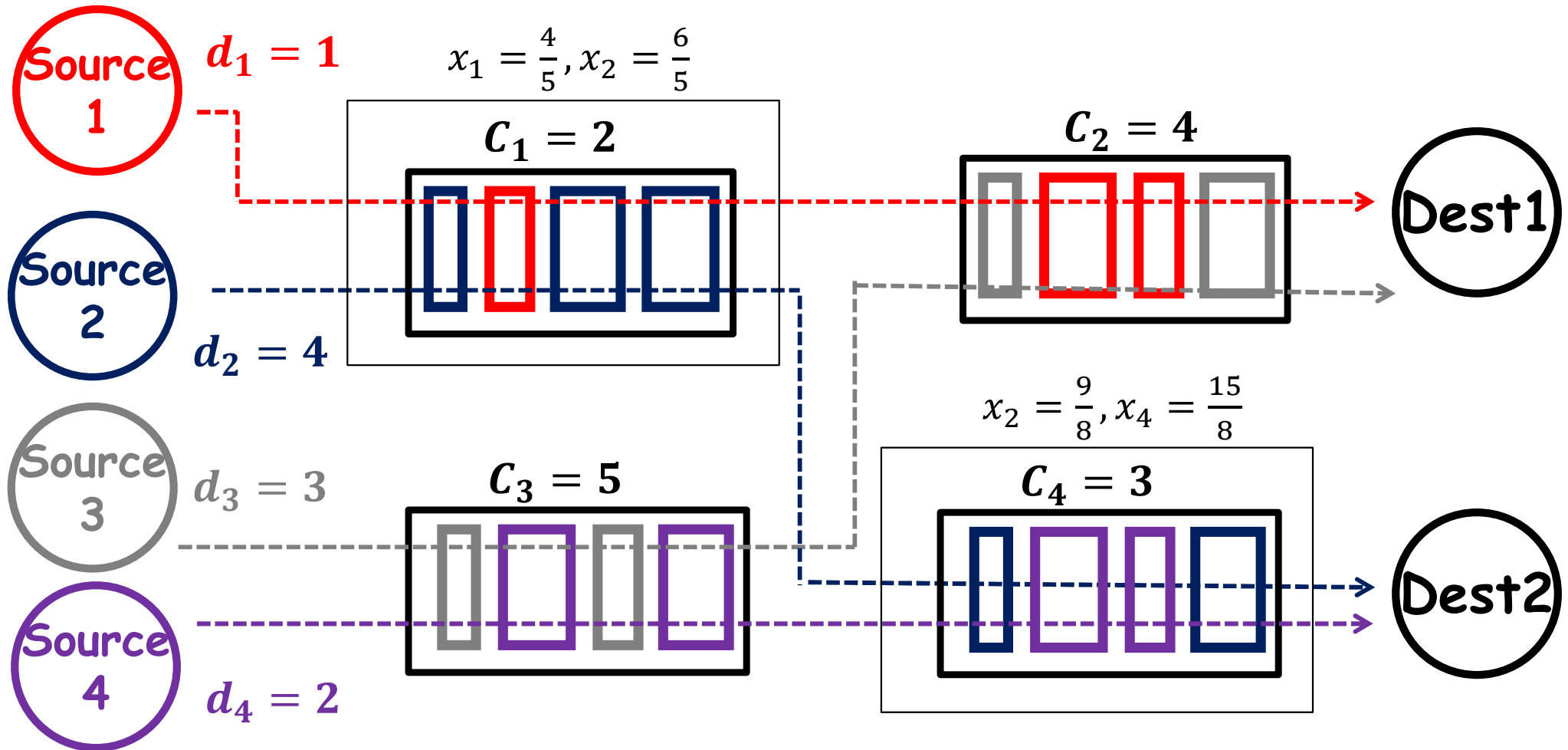
□ Calculate the max-min fair solution:

$$\phi = (\phi_1, \phi_2, \phi_3, \phi_4) = (2, 3, 4, 5)$$

Q2.B

d_1 might be fulfilled by C_1 ($x_1 = 4/5$) and C_2 ($x_1 = 1$)

d_2 can not be fulfilled by C_1 ($x_2 = 6/5$) and C_4 ($x_2 = 9/8$)

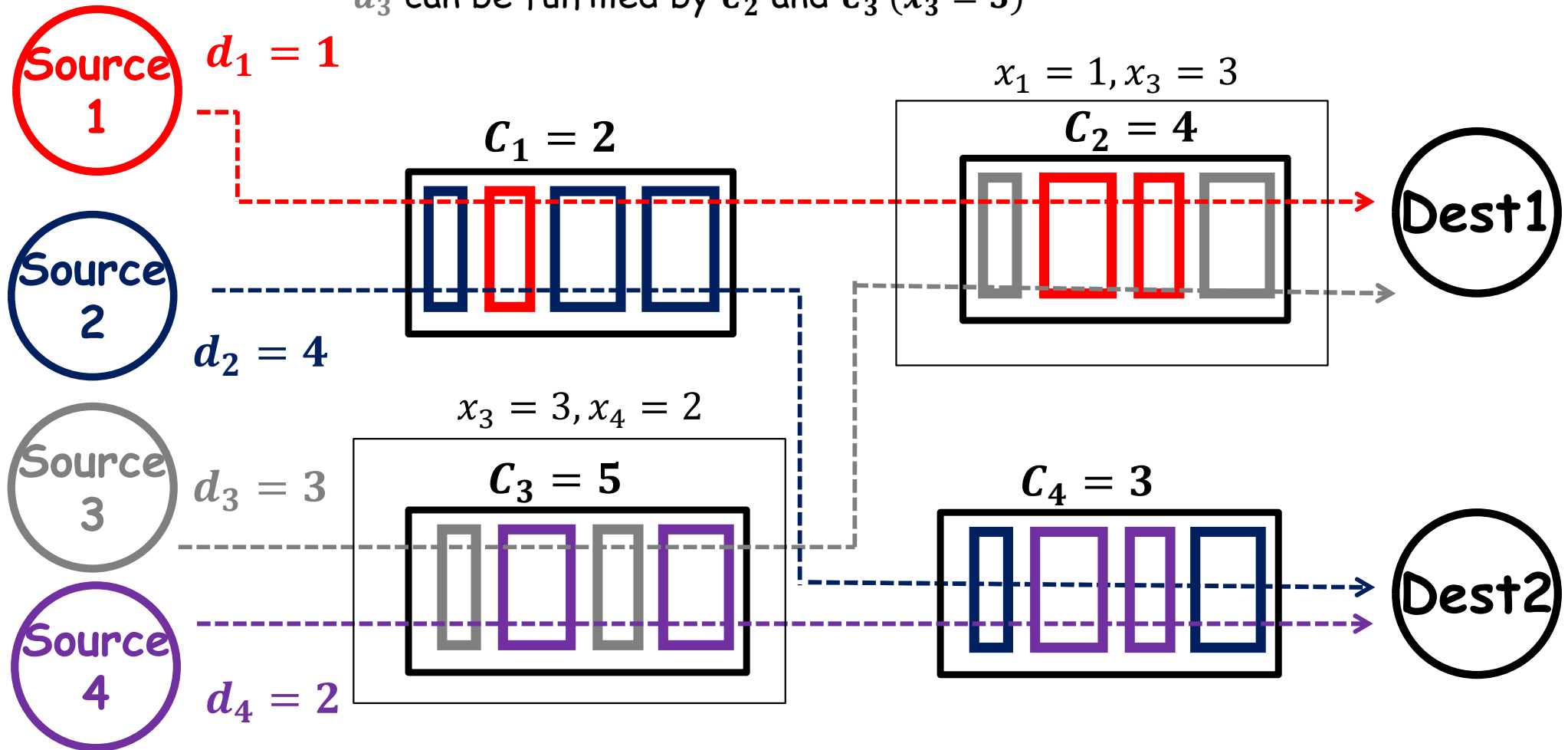


□ Calculate the max-min fair solution:

$$\Phi = (\phi_1, \phi_2, \phi_3, \phi_4) = (2, 3, 4, 5)$$

Q2.B

d_1 might be fulfilled by C_1 ($x_1 = 4/5$) and C_2 ($x_1 = 1$)
 d_2 can not be fulfilled by C_1 ($x_2 = 6/5$) and C_4 ($x_2 = 9/8$)
 d_3 can be fulfilled by C_2 and C_3 ($x_3 = 3$)



□ Calculate the max-min fair solution:

$$\Phi = (\phi_1, \phi_2, \phi_3, \phi_4) = (2, 3, 4, 5)$$

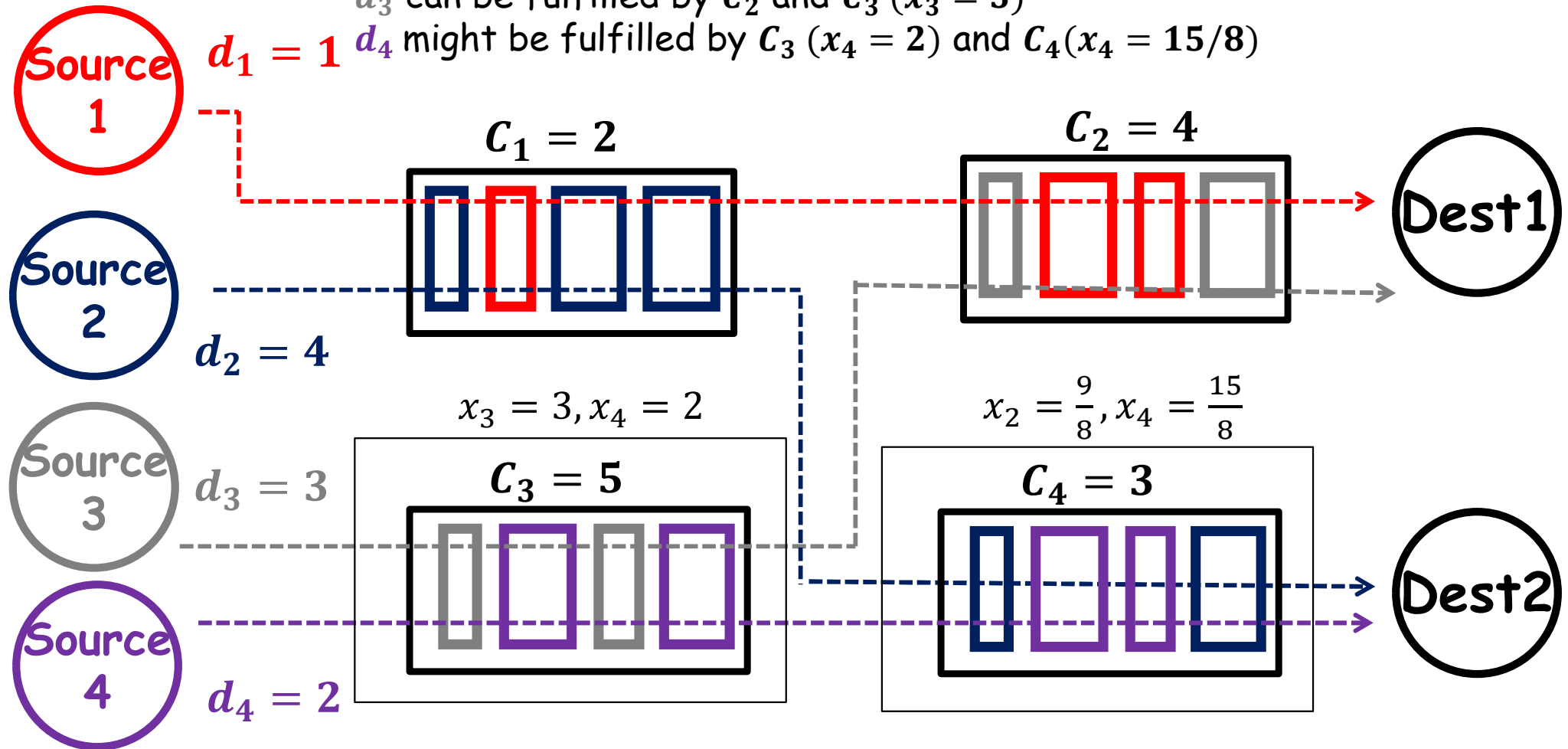
Q2.B

d_1 might be fulfilled by C_1 ($x_1 = 4/5$) and C_2 ($x_1 = 1$)

d_2 can not be fulfilled by C_1 ($x_2 = 6/5$) and C_4 ($x_2 = 9/8$)

d_3 can be fulfilled by C_2 and C_3 ($x_3 = 3$)

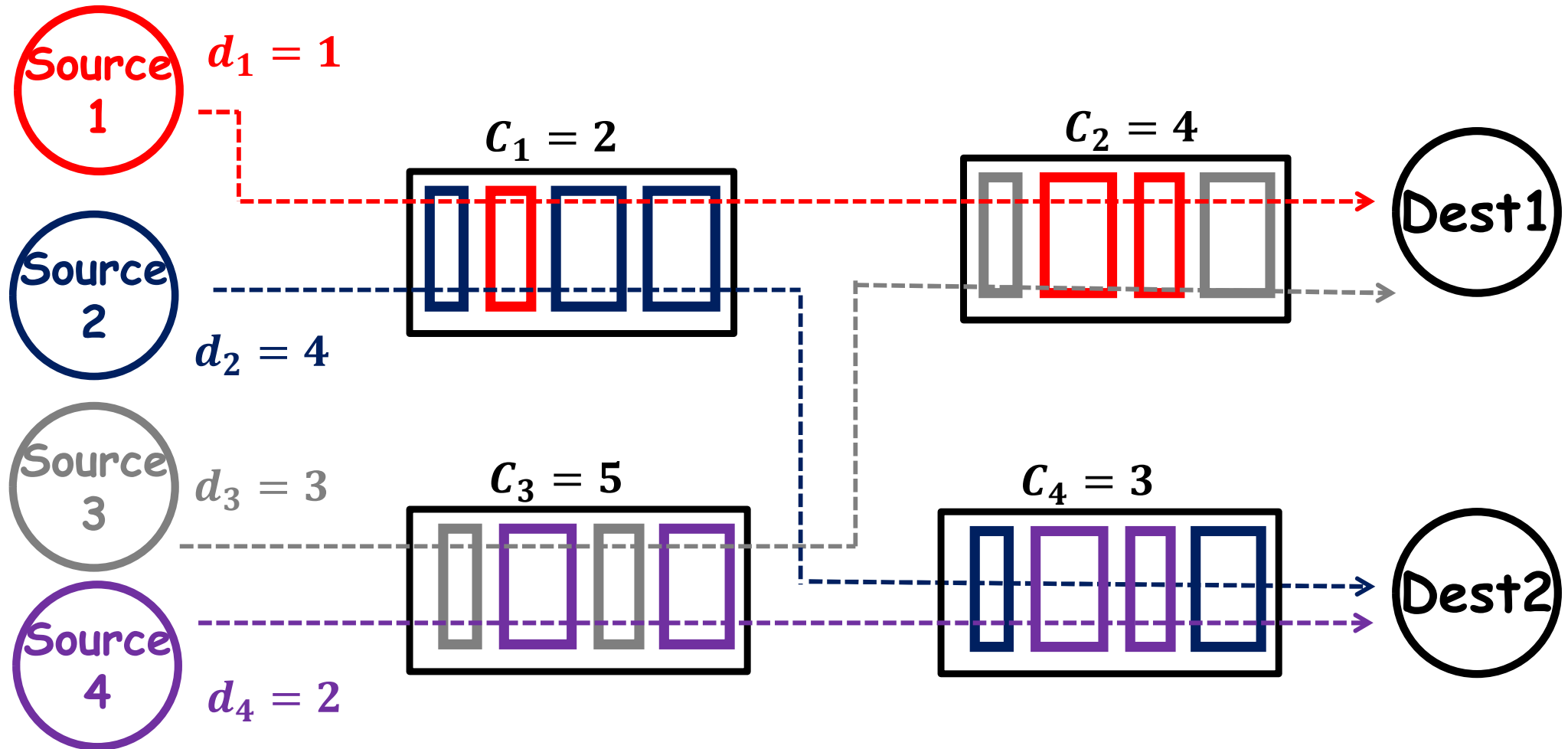
d_4 might be fulfilled by C_3 ($x_4 = 2$) and C_4 ($x_4 = 15/8$)



□ Calculate the max-min fair solution:

$$\Phi = (\phi_1, \phi_2, \phi_3, \phi_4) = (2, 3, 4, 5)$$

Q2.B

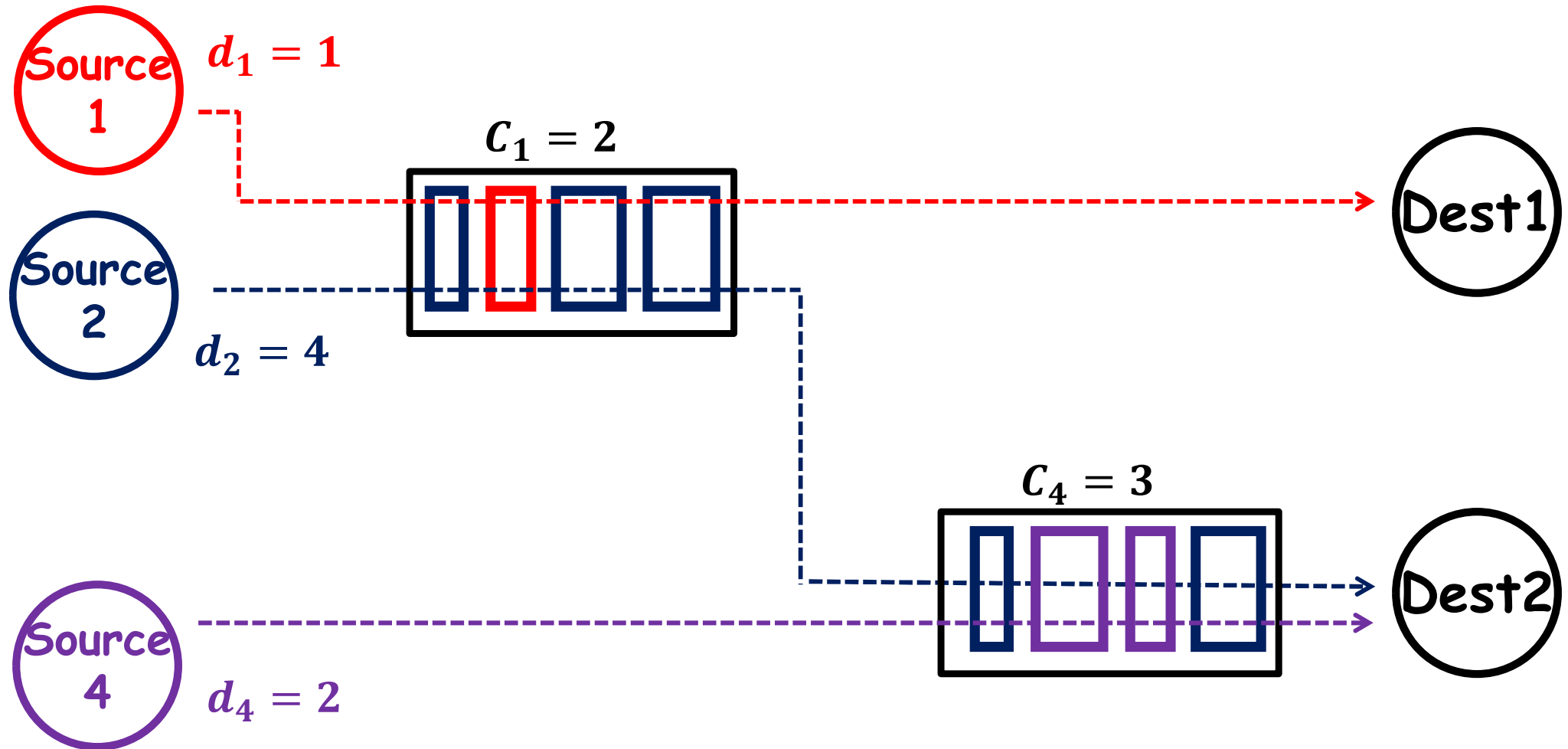


□ Calculate the max-min fair solution:

❖ d_3 can be fulfilled. $\rightarrow x_3 = 3$

$$\Phi = (\phi_1, \phi_2, \phi_3, \phi_4) = (2, 3, 4, 5)$$

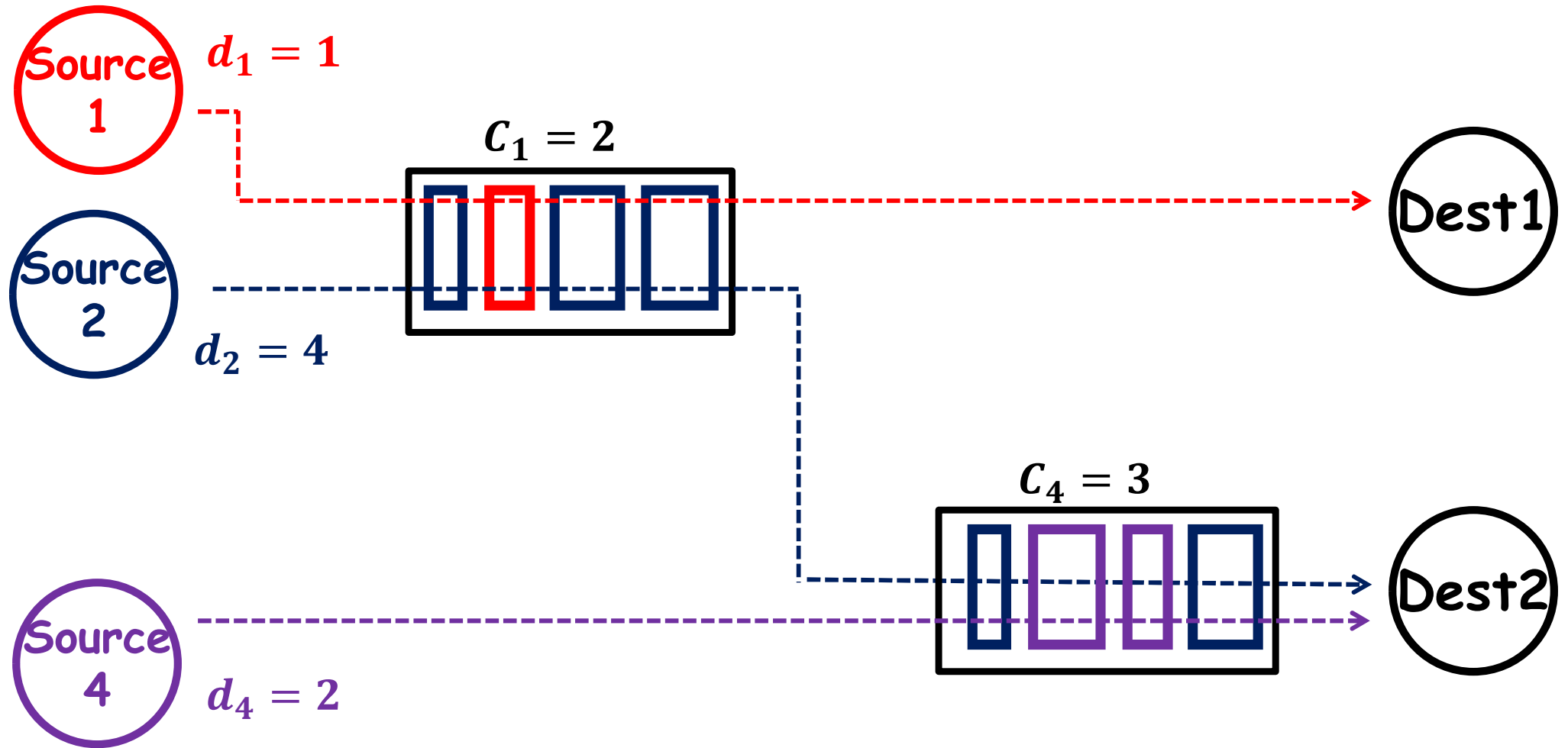
Q2.B



- ❖ Proportional share at C_1 at $(x_1, x_2) = (4/5, 6/5)$
- ❖ Proportional share at C_4 at $(x_2, x_4) = (9/8, 15/8)$
- ❖ $x_2 = \min\{6/5, 9/8\} = 9/8$

$$\Phi = (\phi_1, \phi_2, \phi_3, \phi_4) = (2, 3, 4, 5)$$

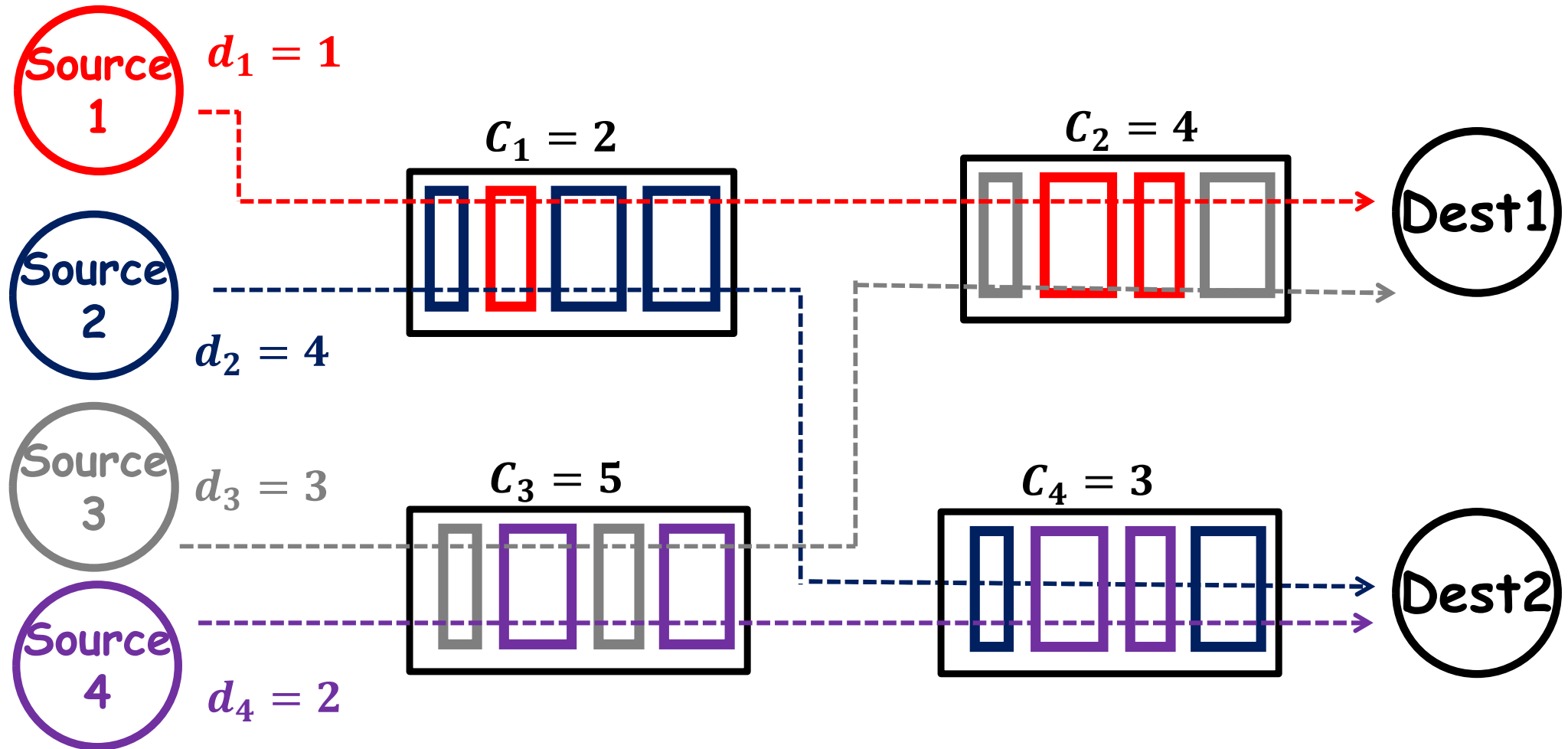
Q2.B



$$x = (x_1, x_2, x_3, x_4) = \left(\frac{7}{8}, \frac{9}{8}, 3, \frac{15}{8} \right)$$

$$\Phi = (\phi_1, \phi_2, \phi_3, \phi_4) = (2, 3, 4, 5)$$

Q2.B



□ Bottleneck links for each flow:

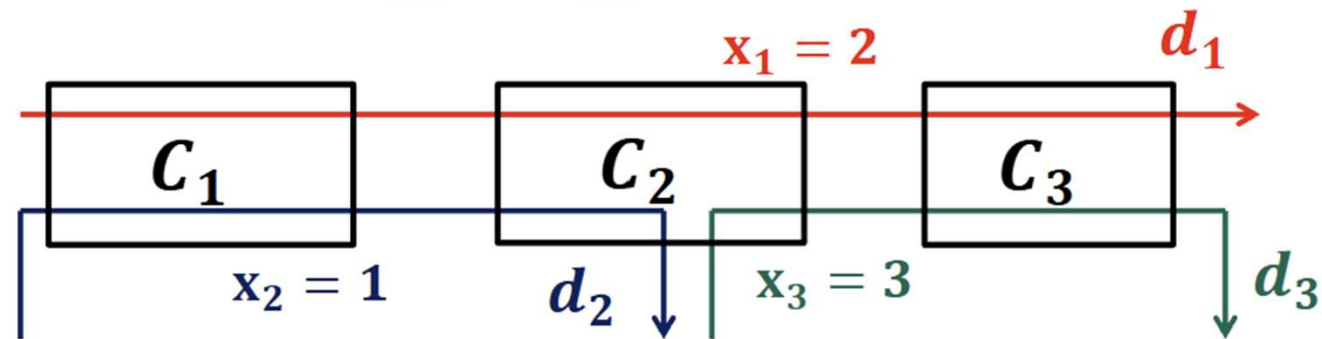
Bottleneck in weighted version

- In the network case, we say that resource r is a bottleneck for flow i if and only if
 - resource r is saturated and
 - flow i has the maximum rate among all flows using resource r .

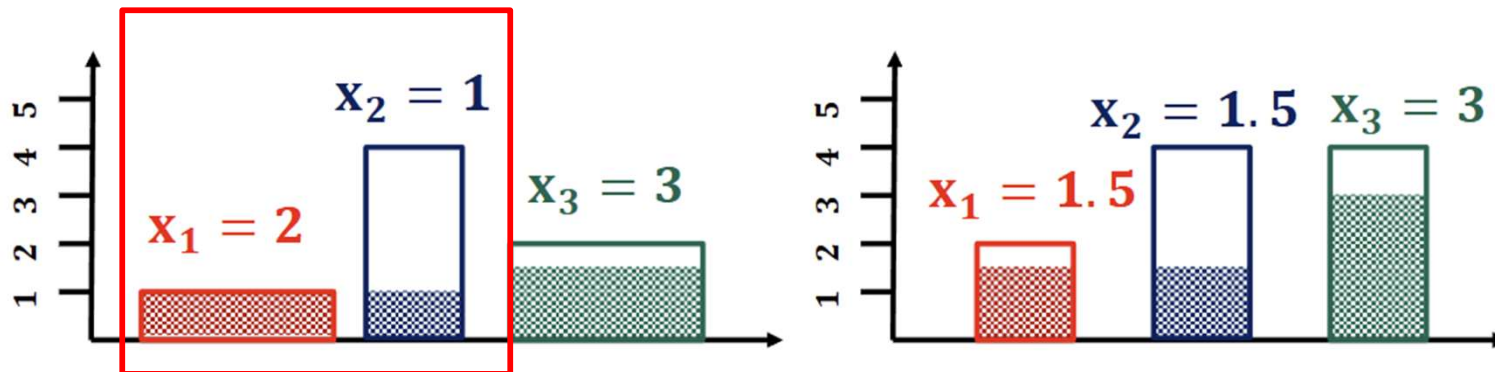


Compare the normalized allocation. $\frac{x_i}{\phi_i}$

Water Filling Algorithm



□ $\mathbf{C} = (3, 6, 7)$, $\mathbf{d} = (2, 4, 4)$ and $\boldsymbol{\phi} = (2, 1, 2)$

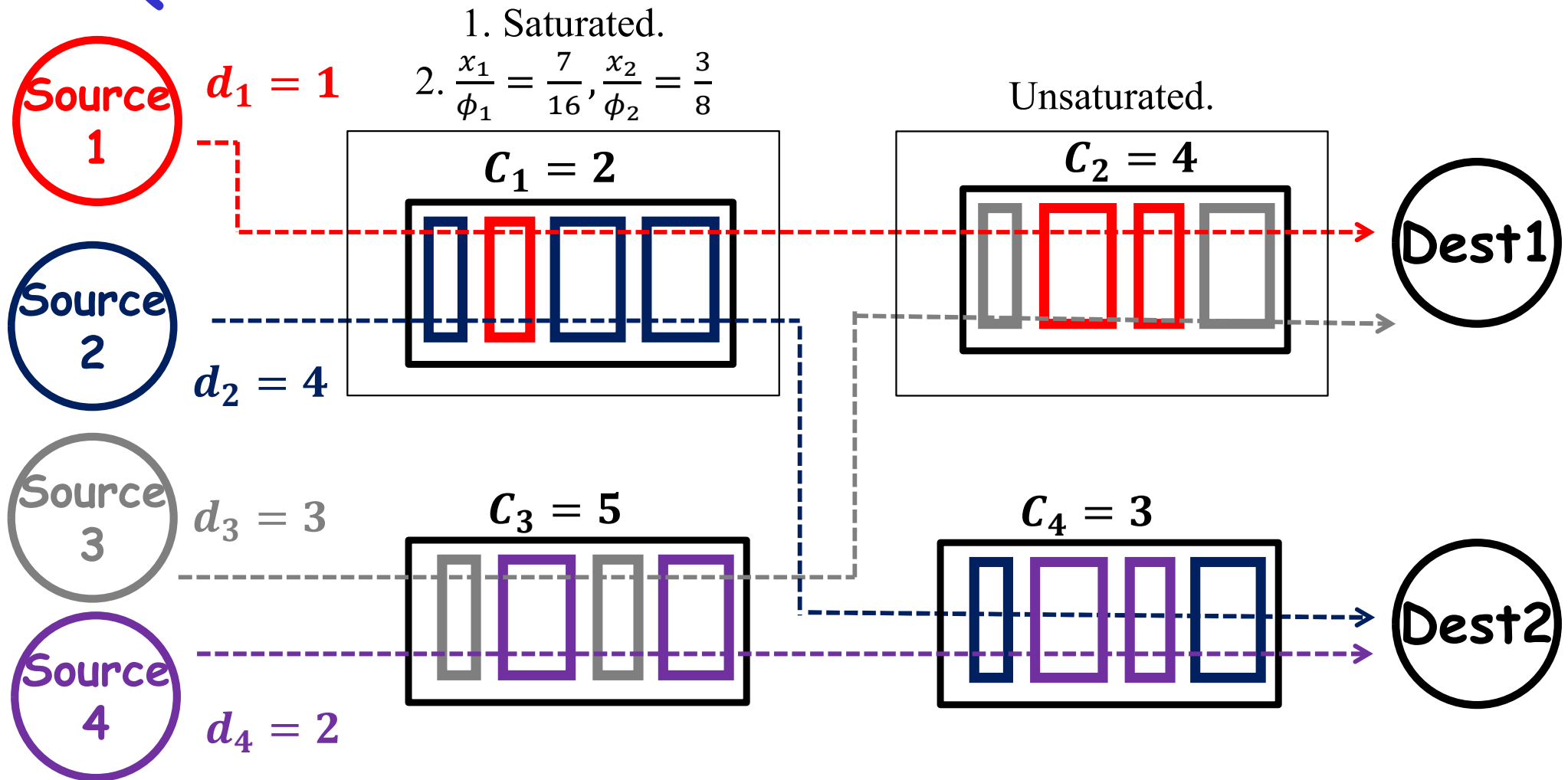


□ Buckets with width ϕ_i and volume d_i

□ Max-min solution for $\boldsymbol{\phi} = (1, 1, 1)$

$$\Phi = (\phi_1, \phi_2, \phi_3, \phi_4) = (2, 3, 4, 5)$$

Q2.B

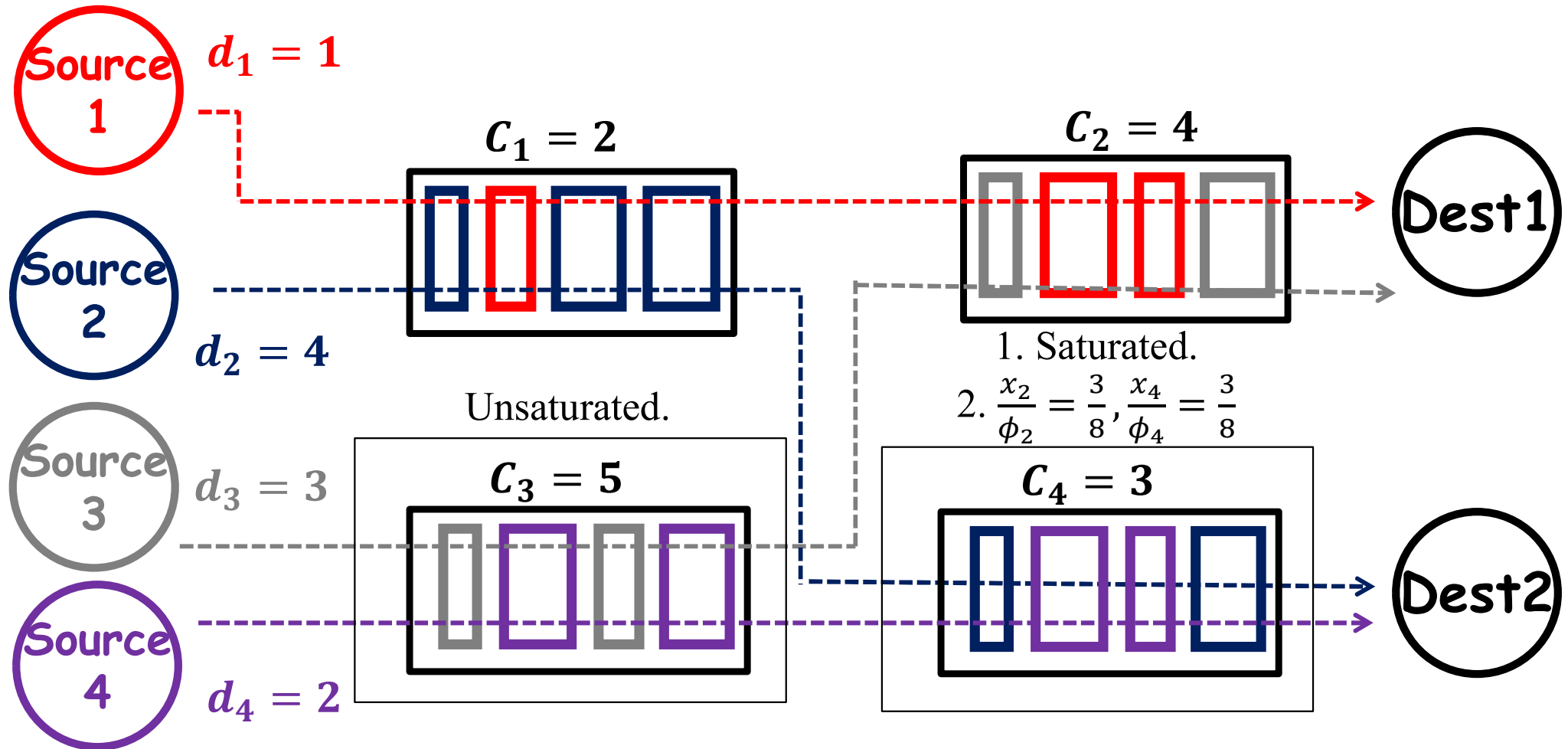


□ Bottleneck links for each flow:

d_1 is bottlenecked by C_1

$$\Phi = (\phi_1, \phi_2, \phi_3, \phi_4) = (2, 3, 4, 5)$$

Q2.B

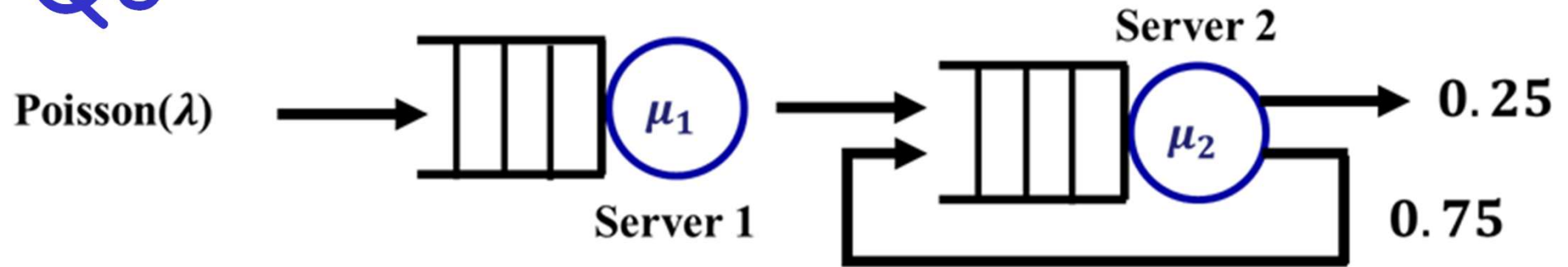


□ Bottleneck links for each flow:

d_1 is bottlenecked by C_1

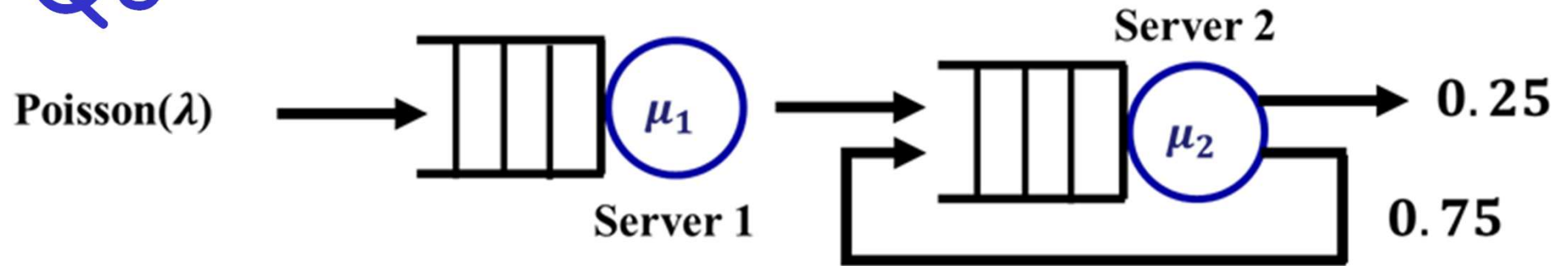
d_2, d_4 is bottlenecked by C_4

Q3



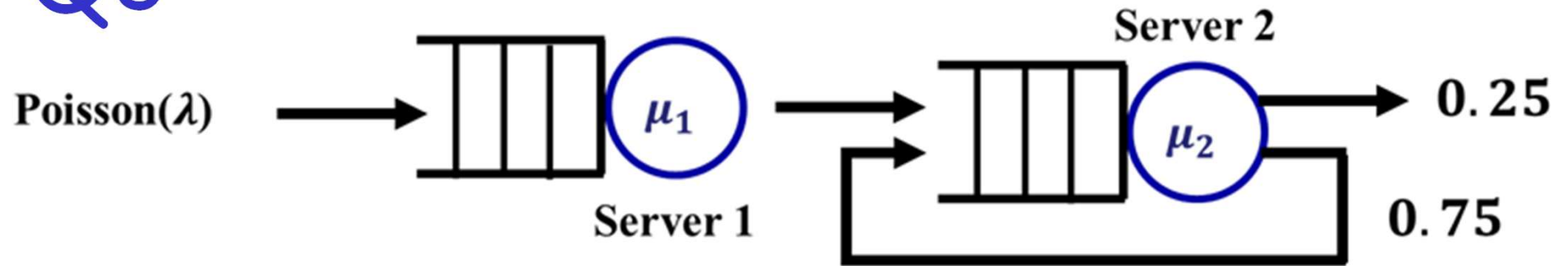
- In the above Jackson network,
 - ❖ how much capacities of μ_1 and μ_2 do we need to guarantee the stability of the system?
 - ❖ suppose we have a total capacity of $\mu = \mu_1 + \mu_2$, how can we allocate the resource such that the average sojourn time of the jobs is minimized?

Q3



$$\begin{cases} \lambda_1 = \lambda \\ \lambda_2 = \lambda_1 + 0.75\lambda_2 \end{cases} \Rightarrow \begin{cases} \lambda_1 = \lambda \\ \lambda_2 = 4\lambda \end{cases} \Rightarrow \begin{cases} \mu_1 > \lambda \\ \mu_2 > 4\lambda \end{cases}$$

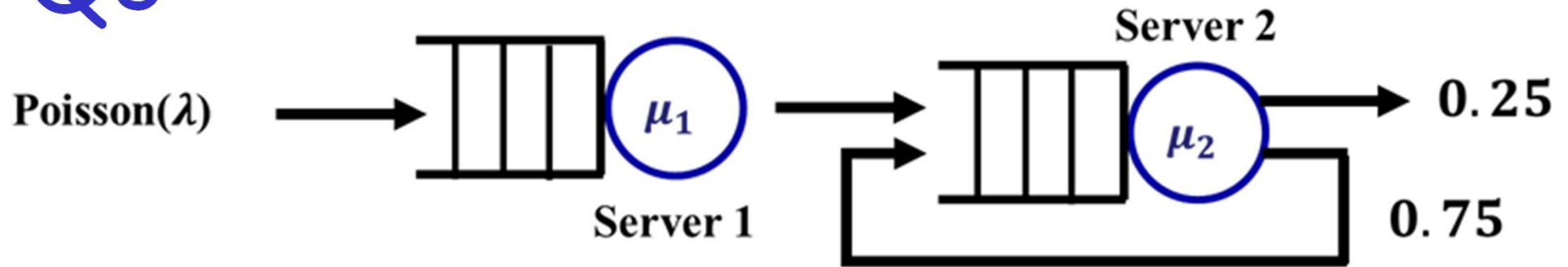
Q3



$$\begin{cases} \lambda_1 = \lambda \\ \lambda_2 = \lambda_1 + 0.75\lambda_2 \end{cases} \Rightarrow \begin{cases} \lambda_1 = \lambda \\ \lambda_2 = 4\lambda \end{cases} \Rightarrow \begin{cases} \mu_1 > \lambda \\ \mu_2 > 4\lambda \end{cases}$$

❖ how to minimize average sojourn time?

Q3



$$\begin{cases} \lambda_1 = \lambda \\ \lambda_2 = \lambda_1 + 0.75\lambda_2 \end{cases} \Rightarrow \begin{cases} \lambda_1 = \lambda \\ \lambda_2 = 4\lambda \end{cases} \Rightarrow \begin{cases} \mu_1 > \lambda \\ \mu_2 > 4\lambda \end{cases}$$

❖ how to minimize average sojourn time?

$$\begin{aligned} \text{minimize } E[W] &= \frac{E[L]}{\lambda} = \frac{E[L_1] + E[L_2]}{\lambda} \\ &= \frac{1}{\lambda} \left(\frac{\rho_1}{1 - \rho_1} + \frac{\rho_2}{1 - \rho_2} \right) \\ &= \frac{1}{\lambda} \left(\frac{\lambda_1}{\mu_1 - \lambda_1} + \frac{\lambda_2}{\mu_2 - \lambda_2} \right) = \frac{1}{\mu_1 - \lambda} + \frac{4}{\mu_2 - 4\lambda} \end{aligned}$$

Since $\rho = \frac{\lambda}{\mu}$

$$\text{minimize } \frac{1}{\mu_1 - \lambda} + \frac{4}{\mu_2 - 4\lambda}; \text{ subject to: } \begin{cases} \mu_1 > \lambda \\ \mu_2 > 4\lambda \\ \mu_1 + \mu_2 = \mu \end{cases}$$

$$\text{minimize } \frac{1}{\mu_1 - \lambda} + \frac{4}{\mu_2 - 4\lambda}; \text{ subject to: } \begin{cases} \mu_1 > \lambda \\ \mu_2 > 4\lambda \\ \mu_1 + \mu_2 = \mu \end{cases}$$

$$\text{let } x = \mu_1, \text{ then } \begin{cases} x > \lambda \\ \mu - x > 4\lambda \end{cases}$$

$$\Rightarrow \begin{cases} \text{minimize } f(x) = \frac{1}{x - \lambda} + \frac{4}{\mu - x - 4\lambda} \\ \text{subject to: } \lambda < x < \mu - 4\lambda \end{cases}$$

$$\text{minimize } \frac{1}{\mu_1 - \lambda} + \frac{4}{\mu_2 - 4\lambda}; \text{ subject to: } \begin{cases} \mu_1 > \lambda \\ \mu_2 > 4\lambda \\ \mu_1 + \mu_2 = \mu \end{cases}$$

$$\Rightarrow \begin{cases} \text{minimize } f(x) = \frac{1}{x - \lambda} + \frac{4}{\mu - x - 4\lambda} \\ \text{subject to: } \lambda < x < \mu - 4\lambda \end{cases}$$

$$f'(x) = -\frac{1}{(x - \lambda)^2} + 4\frac{1}{(\mu - x - 4\lambda)^2}$$

$$\text{minimize } \frac{1}{\mu_1 - \lambda} + \frac{4}{\mu_2 - 4\lambda}; \text{ subject to: } \begin{cases} \mu_1 > \lambda \\ \mu_2 > 4\lambda \\ \mu_1 + \mu_2 = \mu \end{cases}$$

$$\Rightarrow \begin{cases} \text{minimize } f(x) = \frac{1}{x - \lambda} + \frac{4}{\mu - x - 4\lambda} \\ \text{subject to: } \lambda < x < \mu - 4\lambda \end{cases}$$

$$f'(x) = -\frac{1}{(x - \lambda)^2} + 4\frac{1}{(\mu - x - 4\lambda)^2}$$

$$\text{Need } f'(x^*) = 0$$

$$\text{minimize } \frac{1}{\mu_1 - \lambda} + \frac{4}{\mu_2 - 4\lambda}; \text{ subject to: } \begin{cases} \mu_1 > \lambda \\ \mu_2 > 4\lambda \\ \mu_1 + \mu_2 = \mu \end{cases}$$

$$\Rightarrow \begin{cases} \text{minimize } f(x) = \frac{1}{x - \lambda} + \frac{4}{\mu - x - 4\lambda} \\ \text{subject to: } \lambda < x < \mu - 4\lambda \end{cases}$$

$$f'(x) = -(x - \lambda)^{-2} + 4(\mu - x - 4\lambda)^{-2}$$

$$f'(x^*) = 0 \Rightarrow (2(\mu - x^* - 4\lambda)^{-1})^2 = ((x^* - \lambda)^{-1})^2$$

$$\text{Since positive} \Rightarrow 2(\mu - x^* - 4\lambda)^{-1} = (x^* - \lambda)^{-1}$$

$$\Rightarrow 2(x^* - \lambda) = \mu - x^* - 4\lambda \Rightarrow 3x^* = \mu - 2\lambda$$

$$\Rightarrow \begin{cases} \mu_1^* = \frac{1}{3}\mu - \frac{2}{3}\lambda \\ \mu_2^* = \frac{2}{3}\mu + \frac{2}{3}\lambda \end{cases}$$

Thank You